

AERO50006 Mathematics 2, Tutorial Sheet 1 (Signals&Systems part)

1. Use Euler's relation to prove each of the following:

i $\sin \theta = \frac{1}{2j}(e^{j\theta} - e^{-j\theta})$

ii $\cos^2 \theta = \frac{1}{2}(1 + \cos 2\theta)$

$$e^{j\theta} = \cos \theta + j \sin \theta$$

$$\text{a) } e^{j\theta} - e^{-j\theta} = [\cos \theta + j \sin \theta] - [\cos(-\theta) + j \sin(-\theta)]$$

$$= (\cos \theta + j \sin \theta) - (\cos \theta - j \sin \theta) = 2j \sin \theta$$

$$\therefore e^{j\theta} - e^{-j\theta} = 2j \sin \theta$$

$$\therefore \sin \theta = \frac{1}{2j} (e^{j\theta} - e^{-j\theta}),$$

$$\text{dii) } e^{j\theta} + e^{-j\theta} = 2 \cos \theta$$

$$\therefore (e^{j\theta} + e^{-j\theta})^2 = 4 \cos^2 \theta$$

$$\& (e^{j\theta} + e^{-j\theta})^2$$

$$= (e^{j\theta})^2 + 2e^{j\theta}e^{-j\theta} + (e^{-j\theta})^2$$

$$= [\cos^2 \theta + 2j \cos \theta \sin \theta - \sin^2 \theta] + 2[\cos^2 \theta + \sin^2 \theta] + [\cos^2 \theta - 2j \cos \theta \sin \theta - \sin^2 \theta]$$

$$= 1 + 2\cos^2 \theta - 2\sin^2 \theta = 2 + 2\cos^2 \theta - (1 - \cos 2\theta)$$

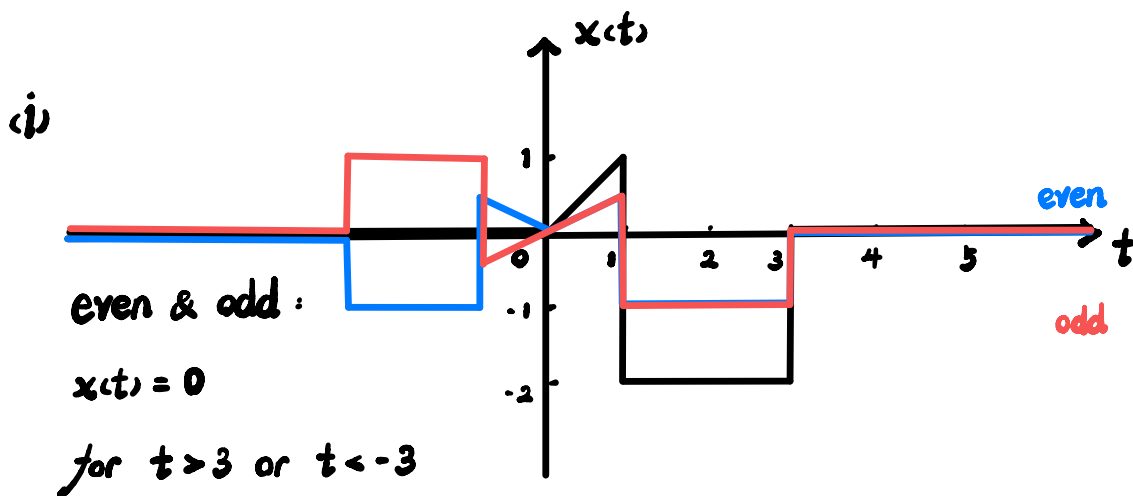
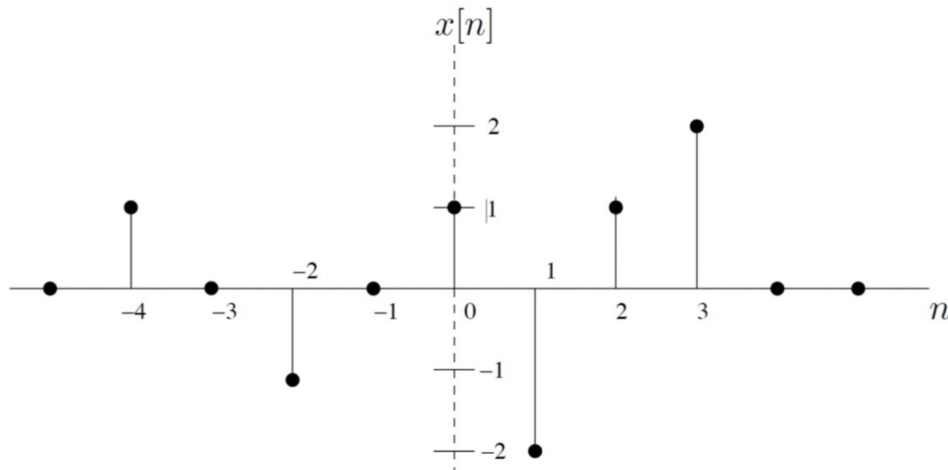
$$\therefore \cos^2 \theta = \frac{1}{2} (1 + \cos 2\theta)$$

2. Find and sketch the even and odd parts of each of the following signals. Also find the energy of each (for a CT signal the energy is defined as $E = \int_{-\infty}^{\infty} |x(t)|^2 dt$ and for a DT signal as $E = \sum_{n=-\infty}^{+\infty} |x[n]|^2$).

(i.)

$$x(t) = \begin{cases} t & 0 < t \leq 1 \\ -2 & 1 < t \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

(ii.) $x[n]$ as shown in the figure below:



$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_0^1 t^2 dt + \int_1^3 |-2|^2 dt$$

$$= \left[\frac{1}{3} t^3 \right]_0^1 + [4t]_1^3$$

$$= \frac{1}{3} + (12 - 4) = \frac{25}{3}$$

(ii) odd : $n = \pm 2$

even & odd : $n > 4$ or $n < -4$

$$E = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

$$= |1|^2 + |0|^2 + |-1|^2 + |0|^2 + |1|^2 + |-2|^2 + |1|^2 + |2|^2$$

$$= 8 + 12$$

3. In the lectures, we defined a linear system as one that is both *scalable* (homogeneous) and *additive*. Determine whether each of the systems below is scalable and/or additive. Justify your answers either with a proof or a counterexample.

(i.)

$$y(t) = \begin{cases} \frac{1}{x(t)} \left[\frac{dx(t)}{dt} \right]^2 & \text{if } x(t) \neq 0 \\ 0 & \text{otherwise} \end{cases}$$

(ii.) $y[n] = \mathcal{R}\{x[n]\}$

(i)

$$x_1 \rightarrow y_1 = \begin{cases} \frac{1}{x_1} \left(\frac{d(x_1)}{dt} \right)^2 & \text{if } x_1 \neq 0 \\ 0 & \text{otherwise} \end{cases}$$

$$x_2 \rightarrow y_2 = \begin{cases} \frac{1}{x_2} \left(\frac{d(x_2)}{dt} \right)^2 & \text{if } x_2 \neq 0 \\ 0 & \text{otherwise} \end{cases}$$

$$\therefore ay_1 \rightarrow \begin{cases} a \left[\frac{1}{x_1} \left(\frac{d(x_1)}{dt} \right)^2 \right] & \text{if } x_1 \neq 0 \\ 0 & \text{otherwise} \end{cases}$$

$$= \begin{cases} \frac{1}{ax_1} \left(\frac{d(ax_1)}{dt} \right)^2 = a \left[\frac{1}{x_1} \left(\frac{d(x_1)}{dt} \right)^2 \right] & \text{if } x_1 \neq 0 \\ 0 & \text{otherwise} \end{cases}$$

$\therefore$ scalable

$$\therefore x_3 = ax_1 + bx_2 \rightarrow$$

$$y_3 = \begin{cases} \frac{1}{x_3} \left(\frac{d(x_3)}{dt} \right)^2 = \frac{1}{ax_1 + bx_2} \left[\frac{d(ax_1 + bx_2)}{dt} \right]^2 & \text{if } x_3 \neq 0 \\ 0 & \text{otherwise} \end{cases}$$

$$\neq ay_1 + by_2 \begin{cases} \frac{a}{x_1} \left(\frac{d(x_1)}{dt} \right)^2 + \frac{b}{x_2} \left(\frac{d(x_2)}{dt} \right)^2 & \text{if } x_1, x_2 \neq 0 \\ 0 & \text{otherwise} \end{cases}$$

$\therefore$ not additive $\therefore$ non-linear

(ii)

$$x_1 \rightarrow y_1 = R\{x_1\}$$

$$\text{e.g. } x(t) = 1 + 2j$$



$$x_2 \rightarrow y_2 = R\{x_2\}$$

not scalable

$$ax_1 \rightarrow ay_1 = aR\{x_1\} = R\{ax_1\} \therefore \text{scalable}$$

$$x_3 = ax_1 + bx_2 \rightarrow$$

$$y_3 = R\{x_3\} = R\{ax_1 + bx_2\} = R\{ax_1\} + R\{bx_2\}$$

$$= ay_1 + by_2 = R\{ax_1\} + R\{bx_2\} \therefore \text{additive}$$

$\therefore$ linear

$$x_1(t) + x_2(t) \xrightarrow{H} R\{x_1(t) + x_2(t)\}$$

$$= R\{x_1(t)\} + R\{x_2(t)\}$$

4. In the lectures, we defined a number of general system properties. In particular, we said that a system can be characterized as one or more of the following:

- A Memoryless
- B Causal
- C Time-invariant
- D Linear
- E Stable

For each of the following systems, determine which of the above properties hold. Justify your answers. For each, assume that $u(t)$ is the input and $y(t)$ is the output.

i $y(t) = \int_{-\infty}^{2t} u(\tau) d\tau,$

ii $y(t) = \sin(x(t)).$

i) A x

B ✓ x

C ✓ x

$$u_1(t) = u(t - \delta)$$

$$\begin{aligned} y_1(t) &= \int_{-\infty}^{2t} u(\tau - \delta) d\tau \stackrel{\xi = \tau - \delta}{=} \int_{-\infty}^{2t} u(\xi) d\tau \\ &= \int_{-\infty}^{2t - \delta} u(\xi) d\xi \neq y(t - \delta) \end{aligned}$$

D ✓

$$ay_1(t) = a \int_{-\infty}^{2t} u_1(\tau) d\tau = \int_{-\infty}^{2t} au_1(\tau) d\tau \therefore \text{scalable}$$

$$u_3(t) = au_1(t) + bu_2(t) \rightarrow$$

$$y_3(t) = \int_{-\infty}^{2t} u_3(\tau) d\tau = \int_{-\infty}^{2t} au_1(\tau) + bu_2(\tau) d\tau$$

$$= a \int_{-\infty}^{2t} u_1(\tau) d\tau + b \int_{-\infty}^{2t} u_2(\tau) d\tau = ay_1(t) + by_2(t) \therefore \text{additive}$$

E x

$$|u(t)| < K$$

$$\begin{aligned} |y(t)| &= \left| \int_{-\infty}^{2t} u(\tau) d\tau \right| = \left| \left[\frac{1}{2} u(\tau) \right]_{-\infty}^{2t} \right| \\ &= \left| \frac{1}{2} u(2t) - \frac{1}{2} u(-\infty) \right| \end{aligned}$$

(ii) A ✓

B ✓

C ✓

$$x_1(t) = x(t-\delta)$$

$$y_1(t) = \sin(x(t-\delta)) = y(t-\delta)$$

D x

$$ay_1 = a \sin(x(t)) \neq \sin(ax(t)) \quad \therefore \text{not scalable}$$

$$x_3 = ax_1 + bx_2 \rightarrow$$

$$y_3 = \sin(x_3(t)) = \sin(ax_1(t) + bx_2(t))$$

$$\neq ay_1 + by_2 = \sin(ax_1(t)) + \sin(bx_2(t)) \quad \therefore \text{not additive}$$

E ✓

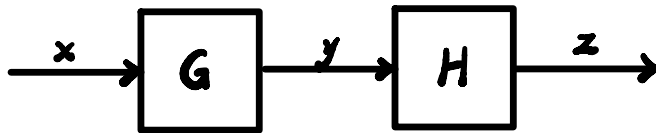
$$|x(t)| \in \mathbb{R}$$

$$|y(t)| = |\sin(x(t))| \quad \therefore -1 < |y(t)| < 1$$

5. Evaluate the truth of each of these statements. If it is true, provide a proof. If not, provide a counterexample.

- i The series interconnection of two linear systems is always linear.
- ii The series interconnection of two nonlinear systems is always nonlinear.

i) **True**



$$G : \sum_{k=0}^N a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^M b_k \frac{d^k x(t)}{dt^k}$$

$$H : \sum_{k=0}^O c_k \frac{d^k z(t)}{dt^k} = \sum_{k=0}^N a_k \frac{d^k y(t)}{dt^k}$$

$$\therefore \sum_{k=0}^O c_k \frac{d^k z(t)}{dt^k} = \sum_{k=0}^M b_k \frac{d^k x(t)}{dt^k}$$

ii) **False**

e.g.

$$G : \sum_{k=-2}^N a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^M b_k \frac{d^k x(t)}{dt^k}$$

$$H : \sum_{k=0}^O c_k \frac{d^k z(t)}{dt^k} = \sum_{k=-2}^N a_k \frac{d^k y(t)}{dt^k}$$

$$\therefore \sum_{k=0}^O c_k \frac{d^k z(t)}{dt^k} = \sum_{k=0}^M b_k \frac{d^k x(t)}{dt^k}$$

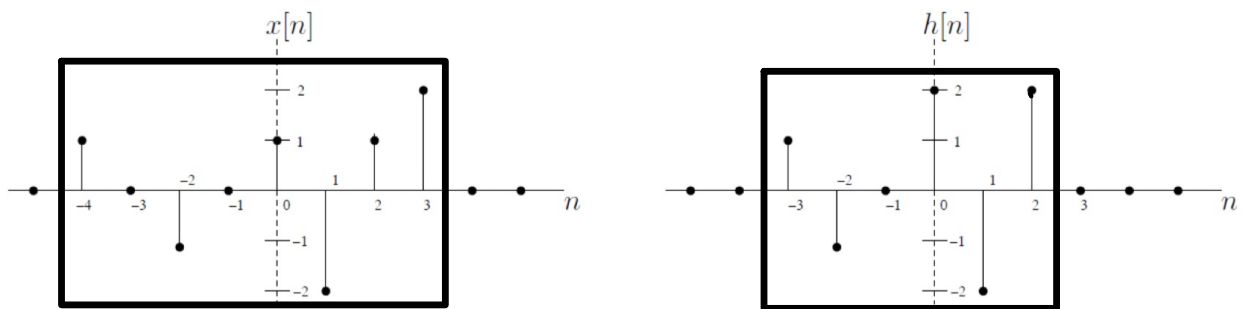
6. For each of the following, calculate and sketch:

$$y[n] = x[n] * h[n].$$

(i.)

$$x[n] = \begin{cases} \left(\frac{1}{3}\right)^n & n \geq 0 \\ 0 & \text{otherwise} \end{cases} \quad h[n] = \begin{cases} -1 & 0 \leq n \leq 10 \\ 0 & \text{otherwise} \end{cases}$$

(ii) $x[n]$ and $h[n]$ as shown below:

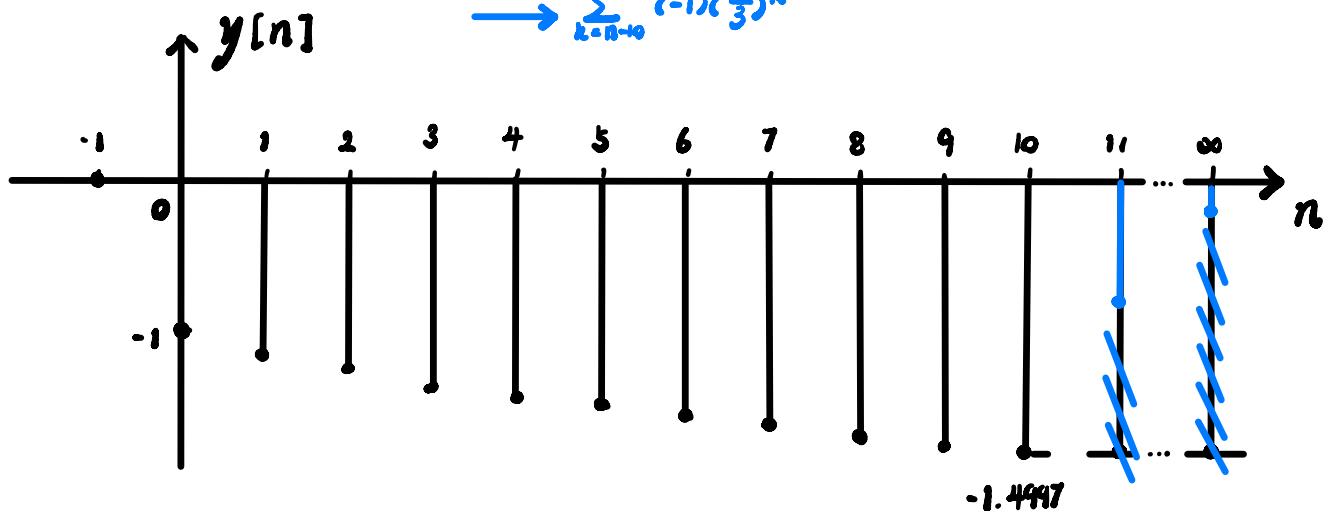


$$(i) \quad y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k]$$

$$\therefore y[n] = \sum_{k=0}^n x[k] h[n-k] \quad \text{for } 0 \leq n \leq 10$$

$$\therefore y[n] = \begin{cases} 0 & n < 0 \\ \sum_{k=0}^n -\left(\frac{1}{3}\right)^k = -\frac{\left(\frac{1}{3}\right)^{n+1} - 1}{\frac{1}{3} - 1} & 0 \leq n \leq 10 \\ \frac{-\left(\frac{1}{3}\right)^{n+1} - 1}{\frac{1}{3} - 1} + \frac{\left(\frac{1}{3}\right)^{n-9} - 1}{\frac{1}{3} - 1} & n > 10 \end{cases}$$

$$\rightarrow \sum_{k=n-10}^n (-1) \left(\frac{1}{3}\right)^k$$

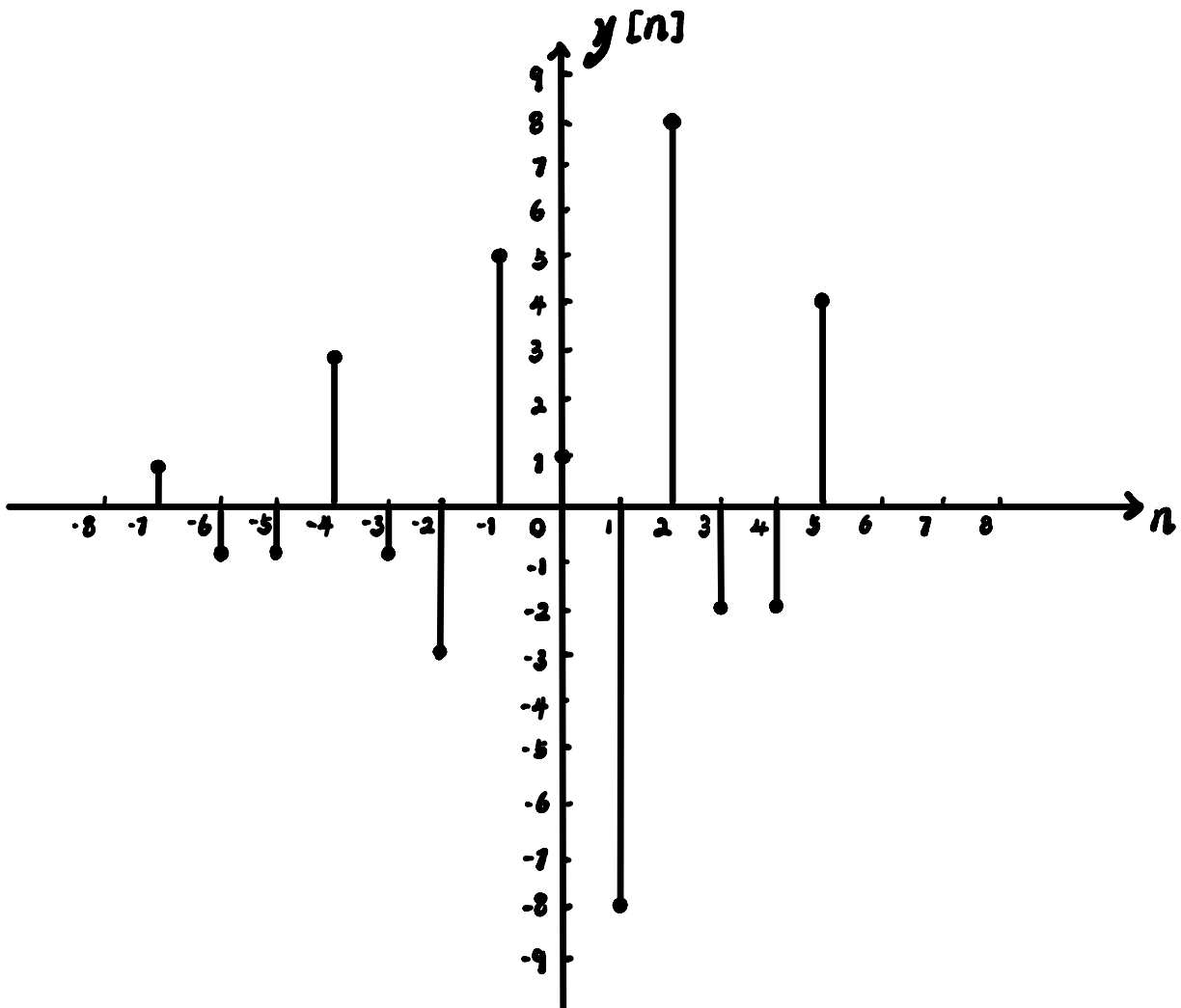


$$\text{ii) } y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

$$\therefore y[n] = \begin{cases} 0 & n < -7 \\ \sum_{k=-4}^n x[k]h[n-k] & -7 \leq n \leq 5 \\ 0 & n > 5 \end{cases}$$

$\therefore$ from -7 to 5.

$$y[n] = 1, -1, -1, 3, -1, -3, 5, 1, -8, 8, -2, -2, 4$$



7. For each of the following, calculate and sketch:

$$y(t) = x(t) * h(t)$$

(a)

$$x(t) = \begin{cases} -1 & -1 \leq t \leq 1 \\ 0 & \text{otherwise} \end{cases} \quad h(t) = \begin{cases} t & 0 \leq t \leq 1 \\ (2-t) & 1 \leq t \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

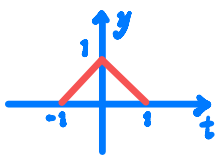
(b)

$$x(t) = \begin{cases} \sin(\pi t) & 0 \leq t \leq 2 \\ 0 & \text{otherwise} \end{cases} \quad h(t) = \begin{cases} 1 & 1 \leq t \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

(c)

$$x(t) = 2 \sum_{k=-\infty}^{\infty} \delta(t - 2k) \quad h(t) = \begin{cases} (1+t) & -1 \leq t \leq 0 \\ (1-t) & 0 \leq t \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

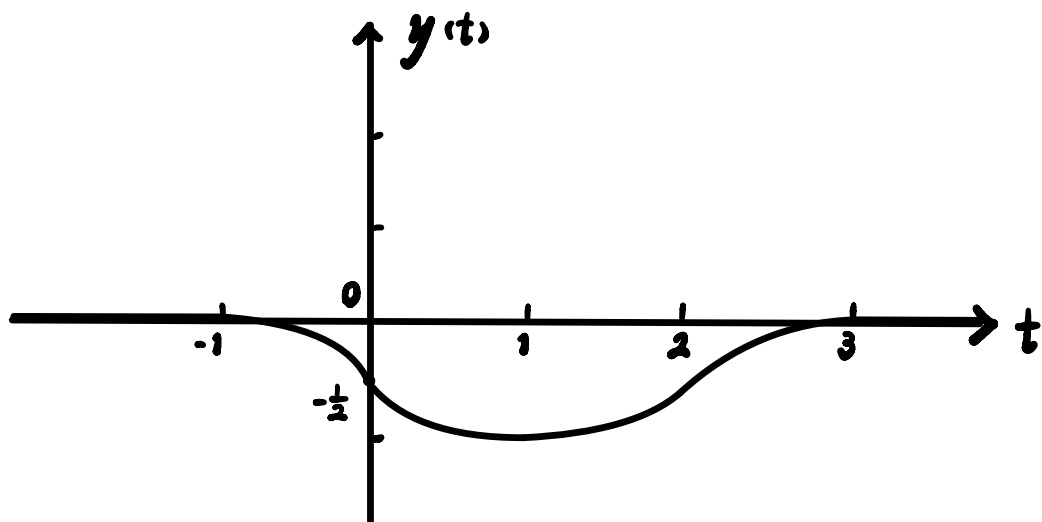
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(a) $y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$

$$\therefore y(t) = \begin{cases} 0 & t \leq -1 \\ \int_{-1}^t \tau - t d\tau & -1 < t \leq 0 \\ \int_{t-1}^t \tau - t d\tau + \int_{-1}^{t-1} -\tau + t - 2 d\tau & 0 < t \leq 1 \\ \int_{t-1}^1 \tau - t d\tau + \int_{t-2}^{t-1} -\tau + t - 2 d\tau & 1 \leq t < 2 \\ \int_{t-2}^1 -\tau + t - 2 d\tau & 2 < t \leq 3 \\ 0 & t > 3 \end{cases}$$

$$\therefore y(t) = \begin{cases} 0 & t \leq -1 \\ -\frac{1}{2}t^2 - t - \frac{1}{2} & -1 < t \leq 0 \\ \frac{1}{2}t^2 - t - \frac{1}{2} & 0 < t \leq 2 \\ -\frac{1}{2}t^2 + 3t - \frac{9}{2} & 2 < t \leq 3 \\ 0 & t > 3 \end{cases}$$



$$b) y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$$

$$\therefore y(t) = \begin{cases} 0 & t \leq 1 \\ \int_0^{t-1} \sin(\pi\tau) d\tau & 1 < t \leq 2 \\ \int_{t-2}^{t-1} \sin(\pi\tau) d\tau & 2 < t \leq 3 \\ \int_{t-2}^2 \sin(\pi\tau) d\tau & 3 < t \leq 4 \\ 0 & t > 4 \end{cases}$$

$$\therefore y(t) = \begin{cases} 0 & t \leq 1 \\ -\frac{1}{\pi} [\cos(\pi(t-1)) - 1] & 1 < t \leq 2 \\ -\frac{1}{\pi} [\cos(\pi(t-1)) - \cos(\pi(t-2))] = \frac{2}{\pi} & 2 < t \leq 3 \\ -\frac{1}{\pi} [1 - \cos(\pi(t-2))] & 3 \leq t < 4 \\ 0 & t > 4 \end{cases}$$

$$(c) \quad x(t) = 2 \sum_{k=-\infty}^{\infty} \delta(t-2k)$$

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$$

$$= \int_0^1 2 \sum_{k=-\infty}^{\infty} \delta(\tau-2k) (1+t-\tau) d\tau \\ + \int_{-1}^0 2 \sum_{k=-\infty}^{\infty} \delta(\tau-2k) (1-t+\tau) d\tau$$

$$\int_{-\infty}^{\infty} h(\tau) \delta(\tau-2k) d\tau = h(2k)$$

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau = 2 \sum_{k=-\infty}^{\infty} \int_{-\infty}^{\infty} \delta(\tau-2k) h(t-\tau) d\tau$$

$$= 2 \sum_{k=-\infty}^{\infty} h(t-2k)$$

$$= \begin{cases} 2 \sum_{k=-\infty}^{\infty} (1+t-2k) & -1 \leq t-2k \leq 0 \\ 2 \sum_{k=-\infty}^{\infty} (1-t+2k) & 0 \leq t-2k \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

8. Suppose that a continuous-time LTI system has impulse response $h(t)$. If the input to this system is a unit step, show that the output is

$$y(t) = \int_{-\infty}^t h(\tau) d\tau.$$

This signal is called the *step response* of the system. **NB:** Like the impulse response, the step response can be used to completely characterize the behaviour of the system with respect to any other input.

$$y(t) = x(t) * h(t) = h(t) * x(t) = \int_{-\infty}^{\infty} h(\tau) x(t-\tau) d\tau$$

$$x(t) = \begin{cases} 0 & t < 0 \\ \frac{t}{\epsilon} & 0 \leq t < \epsilon \\ 1 & t \geq \epsilon \end{cases} \xrightarrow{\epsilon \rightarrow 0} x(t-\tau) = \begin{cases} 1 & \tau \leq t \\ 0 & \tau > t \end{cases}$$

$$y(t) = \begin{cases} \int_{-\infty}^t h(\tau) d\tau & t < 0 \\ \int_{-\infty}^t h(\tau) d\tau & 0 \leq t \leq \epsilon \\ \int_{-\infty}^t h(\tau) d\tau + 0 & t \geq \epsilon \end{cases}$$

$$\therefore y(t) = \int_{-\infty}^t h(\tau) d\tau$$

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$$

$$= \int_{-\infty}^{\infty} x(t-\tau) h(\tau) d\tau$$

$$= \int_{-\infty}^t x(t-\tau) h(\tau) d\tau + \int_t^{\infty} x(t-\tau) h(\tau) d\tau$$

$$= \int_{-\infty}^t h(\tau) d\tau$$

9. Show that if the impulse response of an LTI system satisfies the following condition, then the system is BIBO stable:

$$\int_{-\infty}^{\infty} |h(t)| dt < \infty$$

A system that satisfies the above condition is called *absolutely integrable* (summable for DT systems).

$$\text{If } \int_{-\infty}^{\infty} |h(t)| dt < \infty$$

for unit step.

$$y(t) = x(t) * h(t) = h(t) * x(t) = \int_{-\infty}^{\infty} h(\tau) x(t-\tau) d\tau$$

$$x(t) = \begin{cases} 0 & t < 0 \\ \frac{t}{\epsilon} & 0 \leq t < \epsilon \\ 1 & t \geq \epsilon \end{cases}$$

$$y(t) = \begin{cases} \int_{-\infty}^t h(\tau) d\tau & t < 0 \\ \int_{-\infty}^t h(\tau) d\tau & 0 \leq t \leq \epsilon \\ \int_{-\infty}^t h(\tau) d\tau + 0 & t \geq \epsilon \end{cases}$$

$$\therefore y(t) = \int_{-\infty}^t h(\tau) d\tau$$

$$\therefore |y(t)| < \infty \quad \therefore \text{BIBO stable}$$

9. Show that if the impulse response of an LTI system satisfies the following condition, then the system is BIBO stable:

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A system that satisfies the above condition is called *absolutely integrable* (summable for DT systems).

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$$

$$= \int_{-\infty}^{\infty} x(t-\tau) h(\tau) d\tau$$

$$\begin{aligned} \therefore |y(t)| &= \left| \int_{-\infty}^{\infty} x(t-\tau) h(\tau) d\tau \right| \leq \int_{-\infty}^{\infty} |x(t-\tau) h(\tau)| d\tau \\ &\leq \int_{-\infty}^{\infty} |x(t-\tau)| |h(\tau)| d\tau \end{aligned}$$

$$\text{let } |x(t-\tau)| \leq M$$

$$\therefore |y(t)| \leq \int_{-\infty}^{\infty} M |h(\tau)| d\tau$$

$$\therefore |y(t)| \leq M \int_{-\infty}^{\infty} |h(\tau)| d\tau$$

$$\therefore \int_{-\infty}^{\infty} |h(\tau)| d\tau < \infty$$

$\therefore$ BIBO stable